United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Exam Name: Final Term Exam   Trimester: Summer 2025
Course Code: PMG 4101, Course Title: Project Management
Total Marks: **40**   Duration: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions.

Consider the following Scenario:

Dhaka city is facing challenges with inefficient Waste Management, which has led to increased operational costs, poor recycling rates, and environmental pollution. To address these issues, both Dhaka North & South City Corporation plans to implement a ***Smart Waste Management System*** aimed at optimizing waste collection, reducing landfill overflow, and enhancing recycling processes. Your organization has been selected to design, develop, and implement this robust system, which must handle millions of users across the city, including city residents, waste collection operators and government administrators. The project team consists of 15 experienced software engineers, while 25 freshers have been hired to assist with the large scale of the project. The city corporation's expects regular milestone updates and progress reviews throughout the project.

As the project manager, you are tasked with leading this critical initiative. The system will consist of IoT-enabled waste bins, a real-time data analytics platform, and a mobile application. The smart bins will feature sensors that measure fill levels, weight, and the type of waste (e.g., recyclable, general waste, organic). These bins will notify the centralized platform when they are full, ensuring waste collection trucks are dispatched only when needed, thus optimizing fuel consumption and reducing road congestion. The cloud-based platform will aggregate data from the sensors and analyze it to generate insights for waste collection optimization, such as identifying patterns in waste generation and adjusting collection routes. The platform will also store historical data to help city officials plan long-term waste management strategies and generate reports on recycling rates, waste collection efficiency, and other key performance indicators (KPIs). Residents will use a mobile app to track the status of their local waste bins, receive notifications about collection schedules, and access recycling guidelines. Waste collectors will also use the app to receive real-time updates on collection routes and system malfunctions. The trucks will be equipped with GPS to enable route optimization based on real-time data and external factors like traffic and weather conditions. The system will also allow operators to manually adjust routes in emergencies. Furthermore, privacy and data security concerns must be addressed to ensure compliance with regulations.

After analysis, you have found that all the user interface features [input & output] are relatively average to build but handling of the internal and external data files and query are very complex to build. All the technical complexity factors from F1 to F5 are moderate, F6 to F9 are essential, F12 and F14 are significant and the remaining technical complexity factors are average. The average line of code required per function point is 70.

1. Consider the above scenario, calculate the CT (count total) and Complexity Adjustment Factor (CAF) for this project. Calculate FP (Function point) of this project using the CT and CAF. **[CO3] 4+2+2 = 8**

2. Determine the project complexity according to COCOMO model. Find all the COCOMO estimating parameters for the project mentioned in the above scenario [show all calculations i.e., efforts, duration, budget etc.]. [**CO3**] **1+4 = 5**

3. Explain how change control board (CCB) is formed for software change control perspective comprising different stakeholders of software development group. Write down the purposes, basic courses of events, and alternate paths of Change Control process. **[CO4] 2+4 = 6**

4. Develop a Gantt-chart for the above project scenario by identifying the tasks and tasks' dependency. Also, mention different assumptions or considerations that need to be considered (e.g., weekly holidays and government holidays) for making a pragmatic estimation. **[CO5] 4+2 = 6**

5. Which Review method will be appropriate for reviewing the project above, mention the justification. Describe the activities of that review process. **[CO4] 2+3 = 5**

6. Shortly define KPA concept and it's purposes. As a project manager if you want to achieve the Level IV - CMMI Maturity Level, what are the compliance/rubrics/guidelines you need to follow during a software project development? **[CO4] 2+3 = 5**

7. Progress report submission and presentation on estimation through COCOMO model and Function point methods for a group project has been evaluated earlier also show the project management activities through the suitable PM Tool. (**Do not answer this question**, **already completed in classroom**). **[CO5] 5**

Parameter	**Simple**	**Avg**	**Complex**
EI	3	4	6
EO	4	5	7
EQ	3	4	6
ILF	7	10	15
EIF	5	7	10

Project Type	**a**	**b**	**c**	**d**
Organic	2.4	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.30